

Numerical Stability of Explicit and Implicit Methods for a Bounded Memristor ODE

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Rationale

Memristors are crucial in neuromorphic computing, but their non-linear dynamics can be difficult to model accurately. The system forms an Ordinary Differential Equation (ODE) because the internal state variable's rate of change is driven by time-dependent electrical excitation. In the HP TiO₂ memristor model, this internal state variable represents the width of the doped region and is physically constrained by the device thickness. If a numerical method uses a time-step that is too large, the computed state may overshoot these physical bounds and produce unrealistic current-voltage behavior. This makes the model a suitable case study for comparing explicit and implicit numerical methods, especially in terms of step-size sensitivity, stability, and preservation of physically meaningful solutions.

Objectives

The primary objective of this mini-project is to compare the behavior of explicit and implicit numerical methods when applied to a bounded, non-linear memristor ODE. Specifically, the project aims to:

1. Formulate the first-order non-linear ODE governing the state variable $w(t)$ of an HP TiO₂ memristor driven by a sinusoidal voltage source.
2. Implement and compare the performance of three numerical integration techniques: Explicit Euler's Method, Fourth-Order Runge-Kutta (RK4), and Implicit Euler's Method.
3. Analyze the resulting Voltage-Current (V-I) pinched hysteresis loops and state trajectories to compare accuracy, step-size sensitivity, and possible boundary overshoot.

Methodology

The project will be executed entirely through numerical simulation using standard programming tools (e.g., Python). The methodology is divided into three phases:

1. **Mathematical Formulation:** The memristor will be modeled using the standard linear ion drift model, where the rate of change of the state variable is defined by

$$M(w) = R_{on} \frac{w}{D} + R_{off} \left(1 - \frac{w}{D}\right)$$
$$i(t) = \frac{v(t)}{M(w)}, \quad v(t) = V_0 \sin(\omega t)$$

$$\boxed{\frac{dw}{dt} = \mu_v \frac{R_{on}}{D^2} \frac{v(t)}{M(w)}}$$

Here, $M(w)$ is the memristance, dw/dt is the state rate, $w(t)$ is the doped-region width, t is time, μ_v is ion mobility, R_{on} is the fully doped resistance, R_{off} is the undoped resistance, D is the device thickness, $v(t)$ is the applied voltage, and $i(t)$ is the resulting current. The state variable is physically constrained by $0 \leq w(t) \leq D$.

2. **Numerical Implementation:** Algorithms for Explicit Euler, RK4, and Implicit Euler will be coded to solve the Initial Value Problem (IVP). Several time-step sizes will be tested, including deliberately coarse steps, to compare the stability and accuracy of each method. For Implicit Euler, the scalar nonlinear update equation will be solved numerically at every time-step.
3. **Evaluation and Analysis:** The state trajectory $w(t)$ and V-I characteristic curves generated by each method will be plotted. The evaluation will focus on identifying unphysical overshoot where the calculated state variable exceeds the physical device thickness, comparing how strongly each method depends on the chosen time-step, and assessing whether the resulting hysteresis loop remains physically meaningful.